

Title: Explainable AI Framework for Employee Attrition Prediction

Abstract

Employee attrition is one of the major challenges faced by organizations, as high turnover leads to increased recruitment costs, reduced productivity, and loss of experienced talent. Traditional machine learning models can predict employee attrition with high accuracy but often operate as black-box systems, making it difficult for human resource professionals to understand the reasons behind the predictions. This project proposes an **Explainable AI Framework for Employee Attrition Prediction** that combines machine learning algorithms with Explainable Artificial Intelligence (XAI) techniques to accurately predict employee attrition while providing transparent and interpretable explanations for each prediction.

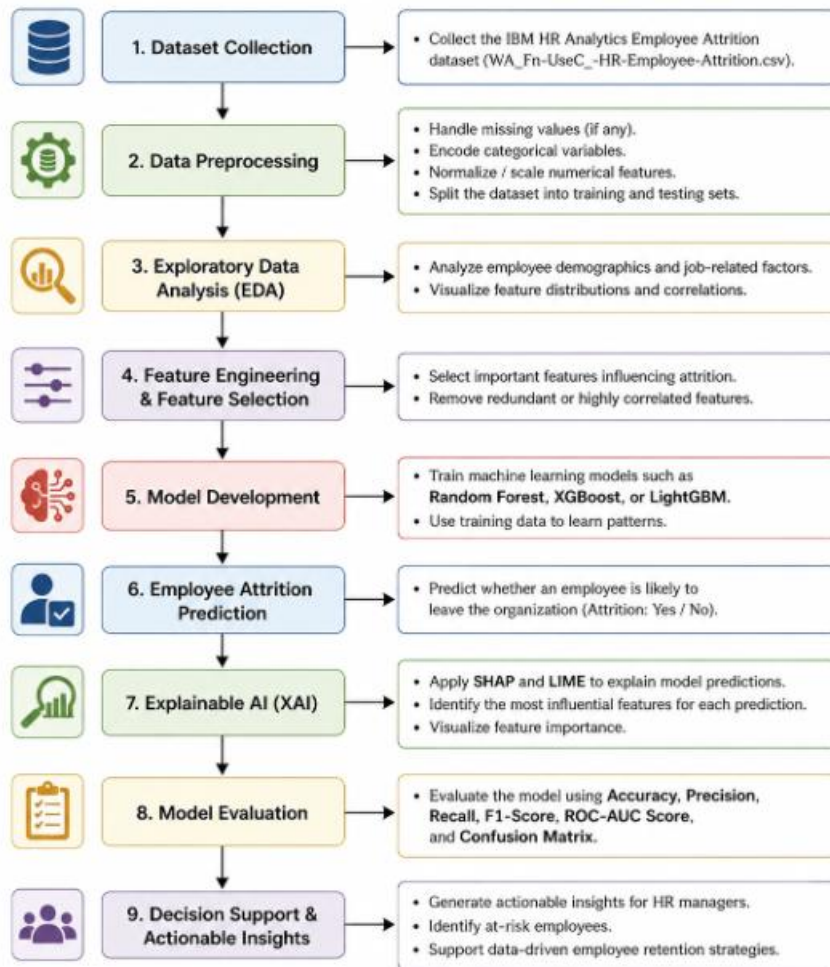
The proposed system utilizes a benchmark employee attrition dataset containing features such as age, job role, monthly income, job satisfaction, work-life balance, overtime, years at the company, and performance rating. The framework performs data preprocessing, feature engineering, data balancing, model training, and employee attrition prediction using machine learning algorithms such as Random Forest, XGBoost, or LightGBM. To enhance interpretability,

Explainable AI techniques such as **SHAP (SHapley Additive exPlanations)**, **LIME (Local Interpretable Model-agnostic Explanations)**, and feature importance analysis are incorporated to identify the key factors influencing attrition predictions and explain individual model decisions.

The proposed framework can assist human resource managers in identifying employees who are at risk of leaving the organization and understanding the primary reasons contributing to attrition, such as job satisfaction, overtime, income, career growth opportunities, and work-life balance. The performance of the prediction model is evaluated using **accuracy, precision, recall, F1-score, ROC-AUC score, and confusion matrix**, while the explainability component is assessed based on the clarity, consistency, and usefulness of the generated explanations. The resulting system aims to provide not only accurate employee attrition prediction but also transparent and actionable insights that support effective workforce planning, employee retention strategies, and data-driven human resource decision-making.

S.No.	Dataset Name	URL Link
1	IBM HR Analytics Employee Attrition & Performance Dataset	https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
2	WA_Fn-UseC_-HR-Employee-Attrition.csv (IBM Dataset)	https://www.ibm.com/communities/analytics/watson-analytics-blog/hr-employee-attrition/
3	HR Analytics Dataset	https://www.kaggle.com/datasets/shantanudhakadd/hr-analytics-dataset
4	HR Analytics Job Change of Data Scientists	https://www.kaggle.com/datasets/arashnic/hr-analytics-job-change-of-data-scientists

APPROACH



Project Group Members

S.No.	Roll.No.	Student Name
1	2420030519	P. Sree bhuvaneswari
2	2420090089	B. Guru prasad
3	2420030759	M. Witty